LECTURE 13

GROUP AND RING THEORY

10. Rational canonical form

The fundamental theorem of finitely generated modules over a PID has interesting applications in obtaining important canonical forms for square matrices over a field. Given an $n \times n$ matrix A over a field F, our goal is to choose an invertible matrix P such that $P^{-1}AP$ is in a desired canonical form.

Not every square matrix is similar to a diagonal matrix. For example, let $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $a \neq 0$, be a matrix over $\mathbb{R}$. Suppose there exists an invertible matrix P such that

$$P^{-1}AP = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}.$$

Clearly,

$$\det(P^{-1}AP - xI) = \det(A - xI).$$

Thus,

$$\begin{vmatrix} d_1 - x & 0 \\ 0 & d_2 - x \end{vmatrix} = \begin{vmatrix} 1 - x & a \\ 0 & 1 - x \end{vmatrix}.$$

This implies that $d_1 d_2 = 1$ and $d_1 + d_2 = 2$, so $d_1 = 1 = d_2$. Hence $P^{-1}AP = I$, equivalently $A = I$, a contradiction.

However, (other) simple canonical forms are important in linear algebra. As an application of the fundamental theorem for finitely generated modules over a PID, we will show that every square matrix A over a field is similar to a matrix

$$\begin{pmatrix} B_1 & & & \\ & B_2 & & \\ & & \ddots & \\ & & & B_s \end{pmatrix}$$

where B_i are matrices of the form

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & * \\ 1 & 0 & \cdots & 0 & * \\ 0 & 1 & \cdots & 0 & * \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & * \end{pmatrix}.$$

Let V be a vector space over a field F, and let $T : V \to V$ be a linear mapping. We can make V an $F[x]$-module by defining the action of any polynomial $f(x) = a_0 + a_1 x + \cdots + a_m x^m$ on any vector $\mathbf{v} \in V$ as

$$f(x)\mathbf{v} = a_0\mathbf{v} + a_1(T\mathbf{v}) + \cdots + a_m(T^m\mathbf{v}),$$

where $T^i\mathbf{v}$ stands for $T^i(\mathbf{v})$. Clearly this action of $F[x]$ on V is the extension of the action of F such that $x\mathbf{v} = T\mathbf{v}$. First we note the following simple fact.

If V is a finite-dimensional vector space over F, then V is a finitely generated $F[x]$-module.

For if $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis of V over F, then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of generators of V as an $F[x]$-module.

Although it follows by Theorem 9.1 that the finitely generated $F[x]$-module V can be decomposed as a finite direct sum of cyclic modules, we provide a proof that gives an explicit algorithm for obtaining the decomposition in this particular situation.

Theorem 10.1. *Let $T \in \mathrm{Hom}_F(V, V)$, and let $f_1(x), \dots, f_n(x)$ be the invariant factors of $A - xI$, where A is a matrix of T. Then*

$$V \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

Proof. (Non-examinable for the oral exam) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis of V, and let $A = (a_{ij})$ be the matrix of T relative to this basis. Then

$$T\mathbf{v}_j = \sum_{i=1}^{n} a_{ij}\mathbf{v}_i, \qquad \text{for } 1 \le j \le n.$$

Let

$$\mathbf{e}_i = (0, \dots, 0, \underbrace{1}_{i\text{th}}, 0, \dots, 0), \qquad \text{for } i = 1, 2, \dots, n,$$

be the standard basis of the $F[x]$-module $F[x]^n = F[x] \times \overset{n}{\cdots} \times F[x]$. As we remarked earlier, we have $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of generators of the $F[x]$-module V. Let

$$\phi : F[x]^n \to V$$

be an $F[x]$-homomorphism sending $\mathbf{e}_i \mapsto \mathbf{v}_i$. We claim

$$\ker \phi = (A - xI)F[x]^n.$$

For, let

$$\begin{pmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{pmatrix} \in F[x]^n,$$

and let $(A - xI)^j$ denote the jth column of $A - xI$. Then

$$\begin{aligned}
&(A-xI)\begin{pmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{pmatrix} \\
&= \begin{pmatrix} (A-xI)^1 & \cdots & (A-xI)^n \end{pmatrix} \begin{pmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{pmatrix} \\
&= (A-xI)^1 g_1(x) + \cdots + (A-xI)^n g_n(x) \\
&= g_1(x)[(a_{11}-x)\mathbf{e}_1 + a_{21}\mathbf{e}_2 + \cdots + a_{n1}\mathbf{e}_n] + \cdots + g_n(x)[a_{1n}\mathbf{e}_1 + \cdots + (a_{nn}-x)\mathbf{e}_n] \\
&\mapsto g_1(x)[(a_{11}-x)\mathbf{v}_1 + \cdots + a_{n1}\mathbf{v}_n] + \cdots + g_n(x)[a_{1n}\mathbf{v}_1 + \cdots + (a_{nn}-x)\mathbf{v}_n] \\
&= g_1(x)[a_{11}\mathbf{v}_1 + \cdots + a_{n1}\mathbf{v}_n - T\mathbf{v}_1] + \cdots + g_n(x)[a_{1n}\mathbf{v}_1 + \cdots + a_{nn}\mathbf{v}_n - T\mathbf{v}_n] \\
&= 0.
\end{aligned}$$

So $(A - xI)F[x]^n \subseteq \ker\phi$. In particular, for all $1 \le j \le n$,

$$\mathbf{w}_j := (A - xI)^j = \sum_{i=1}^n a_{ij}\mathbf{e}_i - x\mathbf{e}_j \in \ker\phi;$$

hence

$$x\mathbf{e}_j = \sum_{i=1}^n a_{ij}\mathbf{e}_i - \mathbf{w}_j. \tag{10.1}$$

Now suppose

$$\sum_{j=1}^n g_j(x)\mathbf{e}_j \in \ker\phi. \tag{10.2}$$

Repeated substitution of $x\mathbf{e}_j$ from (10.1), for $1 \le j \le n$, into (10.2) gives

$$\sum_{j=1}^n g_j(x)\mathbf{e}_j = \sum_{j=1}^n h_j(x)\mathbf{w}_j + \sum_{j=1}^n c_j\mathbf{e}_j,$$

for some $h_j(x) \in F[x]$ and $c_j \in F$. Since $\mathbf{w}_j$ and hence $h_j(x)\mathbf{w}_j \in \ker\phi$ and we have $\sum_{j=1}^n c_j\mathbf{e}_j \in \ker\phi$. Thus $\sum_{j=1}^n c_j\mathbf{v}_j = 0$, which implies that each $c_j = 0$. This proves that

$$\begin{aligned}
\sum_{j=1}^n g_j(x)\mathbf{e}_j &= \sum_{j=1}^n h_j(x)(A - xI)^j \\
&= (A - xI)\begin{pmatrix} h_1(x) \\ \vdots \\ h_n(x) \end{pmatrix} \in (A - xI)F[x]^n.
\end{aligned}$$

Hence $\ker\phi = (A - xI)F[x]^n$, as claimed. Therefore

$$V \cong \frac{F[x]^n}{(A - xI)F[x]^n}.$$

Note that for any $n \times n$ invertible matrix M over $F[x]$, we have $MF[x]^n = F[x]^n$. So, for $n \times n$ invertible matrices P, Q over $F[x]$, we obtain

$$\begin{aligned}
\frac{F[x]^n}{P(A - xI)QF[x]^n} &= \frac{F[x]^n}{P(A - xI)(QF[x]^n)} \\
&= \frac{F[x]^n}{P(A - xI)F[x]^n} \\
&= \frac{P\,F[x]^n}{P(A - xI)F[x]^n} \\
&\cong \frac{F[x]^n}{(A - xI)F[x]^n}
\end{aligned}$$

by a canonical (module) isomorphism (note that multiplying by $r(x)$ is the same as multiplying by $r(x)I$, and this matrix commutes with P) sending

$$P\begin{pmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{pmatrix} + P(A - xI)F[x]^n \longmapsto \begin{pmatrix} g_1(x) \\ \vdots \\ g_n(x) \end{pmatrix} + (A - xI)F[x]^n.$$

Therefore,

$$V \cong \frac{F[x]^n}{(A - xI)F[x]^n} \cong \frac{F[x]^n}{P(A - xI)Q\,F[x]^n},$$

when P and Q are $n \times n$ invertible matrices. Now choose P and Q such that

$$P(A - xI)Q = \operatorname{diag}(f_1(x), \ldots, f_n(x)),$$

where $f_1(x) \mid f_2(x) \mid \cdots \mid f_n(x)$; compare Theorem 8.8. Then

$$\begin{aligned}
V \cong \frac{F[x]^n}{P(A - xI)Q\,F[x]^n} &= \frac{F[x] \oplus \cdots \oplus F[x]}{f_1(x)F[x] \oplus \cdots \oplus f_n(x)F[x]} \\
&\cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.
\end{aligned}$$

□

Remark 10.2. By abuse of terminology $f_1(x), \ldots, f_n(x)$ are also called the invariant factors of A or of T. The sequence of factors $f_k(x), f_{k+1}(x), \ldots, f_n(x)$ obtained from the invariant factors of $A - xI$ by discarding units, if any, are known as the *invariant factors* of the $F[x]$-module V.

Definition 10.3. Let $T \in \operatorname{Hom}_F(V, V)$. If V as an $F[x]$-module relative[1] to T is cyclic, then V is called a *T-cyclic space*. In this case T is also called a *cyclic endomorphism* of V.

Theorem 10.4. *Let W be a submodule of the $F[x]$-module V. Let $T \in \operatorname{Hom}_F(V, V)$ be such that $TW \subseteq W$. Then W is a T-cyclic subspace if and only if there exists an element $w \in W$ such that $\{w, Tw, \ldots, T^{k-1}w\}$ is a basis of W for some $k \in \mathbb{N}$.*

[1] i.e. the action of x on V is given by T

Proof. Let $\dim V = n$, and let W be a T-cyclic subspace generated by w. Since $\dim V = n$, the vectors $w, Tw, T^2w, \dots, T^n w$ must be linearly dependent over F. Let $T^r w$ be the first of these vectors that is linearly dependent on the preceding ones. Then $w, Tw, \dots, T^{r-1}w$ are linear independent, while

$$T^r w = a_0 w + \cdots + a_{r-1}(T^{r-1}w), \qquad \text{for some } a_i \in F. \tag{10.3}$$

We claim that every $T^i w$ is linearly dependent on $w, Tw, \dots, T^{r-1}w$, for all i. For $i \le r-1$ this is clear, and for $i = r$ it follows by (10.3). Thus, let us take $i > r$ and use induction on i. Applying T^{i-r} to (10.3), we obtain

$$T^i w = a_0(T^{i-r}w) + \cdots + a_{r-1}(T^{i-1}w),$$

and by the induction hypothesis each term on the right side is linearly dependent on w, Tw, $\dots$, $T^{r-1}w$. Hence $T^i w$ is also. Since $W = F[x]w$, every element of W is of the form

$$(b_0 + b_1 x + \cdots + b_m x^m)w = (b_0 I + b_1 T + \cdots + b_m T^m)w,$$

where $b_i \in F$. It follows then from these considerations that $\{w, Tw, \dots, T^{r-1}w\}$ is a basis of W.

Conversely, let $\{w, Tw, \dots, T^{r-1}w\}$ be a basis of $F[x]$-module W. This implies that for any element $c_0 w + c_1(Tw) + \cdots + c_{r-1}(T^{r-1}w)$ of W, there exists a polynomial $f(x) = c_0 + c_1 x + \cdots + c_{r-1}x^{r-1}$ such that

$$c_0 w + c_1(Tw) + \cdots + c_{r-1}(T^{r-1}w) = (c_0 + c_1 x + \cdots + c_{r-1}x^{r-1})w.$$

Hence $W \subseteq F[x]w$. Since $F[x]w \subseteq W$, it follows that $W = (w)$. In other words, the subspace W is a T-cyclic subspace. $\square$

Definition 10.5. Let $f(x) = a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} + x^k$ be a monic polynomial over F. The *companion matrix* of $f(x)$ is the $k \times k$ matrix

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix}.$$

Theorem 10.6. *Let $T \in \mathrm{Hom}_F(V, V)$, and let W be a T-cyclic subspace of V with a basis $\{w, Tw, \dots, T^{k-1}w\}$. Then the matrix of T on W is the $k \times k$ companion matrix of a monic polynomial $f(x)$, where $(f(x)) = \mathrm{Ann}(w)$, the annihilator of w in the ring $F[x]$.*

Proof. Since $\{w, Tw, \dots, T^{k-1}w\}$ is a basis of W,

$$T^k w = a_0 w + a_1(Tw) + \cdots + a_{k-1}(T^{k-1}w),$$

where a_i are unique elements of F. Thus,

$$(a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k)w = 0. \tag{10.4}$$

If $\mathrm{Ann}(w) = (f(x))$, then

$$a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k \in (f(x)),$$

so $k \ge \deg f(x) = s$, say. So if $f(x) = b_0 + b_1 x + \cdots + b_{s-1}x^{s-1} + x^s$, then

$$\begin{aligned} 0 &= (b_0 + b_1 x + \cdots + b_{s-1}x^{s-1} + x^s)w \\ &= b_0 w + b_1 T(w) + \cdots b_{s-1}T^{s-1}(w) + T^s w. \end{aligned}$$

Hence $T^s w$ is a linear combination of $w, Tw, \ldots, T^{s-1}w$, whence $s \geq k$. This gives $s = k$. Therefore, if $g(x) := a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k$, then by (10.4) we have $g(x) = cf(x)$, where $c \in F$. By comparing the coefficients of x^k on each side, we get $c = -1$. Hence $f(x) = -g(x)$. As

$$T^k w = a_0 w + a_1(Tw) + \cdots + a_{k-1}(T^{k-1}w),$$

we obtain that the matrix of T on W is

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & a_0 \\ 1 & 0 & \cdots & 0 & a_1 \\ 0 & 1 & \cdots & 0 & a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & a_{k-1} \end{pmatrix},$$

which is the companion matrix of $f(x)$. □

Remark 10.7. In Theorem 10.6, we call $f(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_1 x - a_0$ the minimal polynomial of T on W.